Design and Implementation of a PID-Based Tank Level Control System Using TIA Portal and Factory I/O

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1 – Introduction

1.1 Project Overview

This project focuses on the design and implementation of an automated tank level control system using a Programmable Logic Controller (PLC). The system is developed in TIA Portal and simulated using Factory I/O in conjunction with PLCSIM. The main objective is to regulate the liquid level inside a tank by controlling the opening of a filling valve based on real-time measurements.

A feedback control strategy is employed, where the actual tank level is continuously measured and compared to a desired setpoint. The control action is performed using a PID controller, which adjusts the valve position to minimize the error between the process value and the setpoint. This ensures stable and accurate control of the liquid level under varying conditions.

1.2 Objectives

- To design a complete tank level control system using PLC programming in TIA Portal
- To implement a PID-based control strategy for maintaining the desired liquid level
- To process analog signals using normalization and scaling techniques
- To integrate the PLC with a simulation environment (Factory I/O) using PLCSIM
- To develop control logic including start/stop operations, system enable, and safety interlocks
- To implement basic fault detection and alarm conditions for safe system operation
- To analyze the performance of the control system in terms of stability and accuracy

2 - System Description

2.1 Tank Level Control System Overview

The system under consideration is an industrial liquid-level control system designed to maintain the level of a fluid inside a storage tank at a desired value. The control objective is to ensure that the actual tank level follows a predefined setpoint despite disturbances such as changes in inflow or outflow.

The system operates based on a closed-loop control principle. A level sensor continuously measures the height of the liquid inside the tank and sends this information to the PLC. The controller compares this measured value with the desired setpoint and calculates the appropriate control action using a PID algorithm. This action is then used to adjust the opening of the fill valve, which regulates the inflow of liquid into the tank.

2.2 Working Principle

The operation of the system begins when the user sets a desired level (setpoint). The level sensor continuously measures the actual level of the liquid in the tank and sends this value to the PLC as an analog signal.

Inside the PLC, the measured signal is first processed through normalization and scaling to convert it into engineering units. The scaled process value is then compared with the scaled setpoint. The difference

between these two values, known as the error, is used by the PID controller to determine the required control action.

The controller output adjusts the opening of the fill valve, increasing or decreasing the inflow rate accordingly. If the tank level is below the setpoint, the valve opens more to increase inflow. If the level exceeds the setpoint, the valve closes to reduce inflow.

This feedback loop continues continuously, allowing the system to maintain the liquid level close to the desired value with minimal error and stable behavior.

During simulation, the discharge valve is set to a constant value to represent a fixed outflow from the tank. This introduces a disturbance in the system and allows evaluation of the controller performance under realistic operating conditions.

3 - I/O Configuration

3.1 Overview

The system uses both digital and analog signals to interface between the PLC and the simulated process in Factory I/O. Digital signals are used for control commands and status indicators, while analog signals are used for process measurements and control outputs.

3.2 Digital Inputs and Outputs

3.2.1 Digital Inputs

Tag Name	Address	Description
Start	I0.0	Starts the system operation
Stop	I0.1	Stops the system
Reset_PID	I0.2	Resets the PID controller

3.2.2 Digital Outputs

Tag Name	Address	Description
Start_Light	Q0.0	Indicates system is running
Stop_Light	Q0.1	Indicates system is stopped
Alarm_Light	Q0.2	Indicates fault condition

3.3 Analog Inputs and Outputs

3.3.1 Analog Inputs

Tag Name	Address	Description
Level_Meter	%ID30	Measures tank level (0–10 V)
Flow_Meter	%ID34	Measures inflow rate
Setpoint	%ID38	Desired tank level input

3.3.2 Analog Outputs

Tag Name	Address	Description
Fill_Valve	%QD30	Controls valve opening (0–10 V)
PV_Display	%QD42	Displays actual tank level
SP_Display	%QD38	Displays setpoint value

4 - Control System Design

4.1 Control Strategy

The system uses a closed-loop control strategy to maintain the tank level at a desired setpoint. The measured level is continuously compared with the setpoint, and the controller adjusts the fill valve based on the error between them. This allows the system to automatically correct any deviation and maintain stability.

4.2 PID Control

A PI controller is used to regulate the tank level. The proportional part improves the response speed, while the integral part eliminates steady-state error. The derivative term is not used since the system is slow and does not require it. The controller continuously adjusts the valve to minimize the difference between the actual level and the setpoint.

4.3 Signal Scaling

Analog signals are processed before control. The NORM_X function converts the input from 0–10 V to a normalized value (0–1), and the SCALE_X function converts it to engineering units (0–300 for level). This ensures that the process value and setpoint are in the same range for accurate control.

4.4 Control Loop Operation

The sensor measures the level, the signal is scaled, and the controller compares it with the setpoint. The PI controller then calculates the control output, which is used to adjust the valve opening. This process repeats continuously to maintain a stable tank level.

5 - PLC Program (LAD Logic)

5.1 Program Structure

The PLC program is divided into two main parts. OB1 handles the main control logic such as start/stop and safety conditions, while the cyclic interrupt block (OB123) executes the PID controller at a fixed time interval to ensure stable performance.

5.2 Start/Stop and System Enable

The system is controlled using Start and Stop push buttons. Pressing Start activates the system, while pressing Stop deactivates it. These signals determine whether the controller and outputs are active.

5.3 PID Implementation

The PID controller is implemented using the PID_Compact block inside the cyclic interrupt block. It receives the scaled setpoint (SP) and process value (PV), and calculates the control output used to adjust the fill valve. The controller operates only when the system is enabled to ensure safe operation.

5.4 Output Control

Additionally, when the system is stopped or a fault condition occurs, the valve output is forced to zero to prevent unintended filling and ensure safe operation.

5.5 Reset Logic

A reset function is implemented to clear faults and reinitialize the system. The PID controller can also be reset to ensure proper operation after a fault or restart.

5.6 Safety and Alarms

High and low level limits are implemented using comparator functions. A low level alarm is triggered when the tank level falls below 75, while a high level alarm is triggered when the level exceeds 285. These conditions generate fault signals to prevent unsafe operation. When a fault occurs, the system is stopped and the fill valve is closed.

6 - HMI and User Interface

6.1 HMI Function

The Human Machine Interface (HMI), implemented through the Factory I/O environment, provides basic control and monitoring of the tank level system. It allows the operator to interact with the system using simple input and indicator elements.

The HMI includes three control buttons: Start, Stop, and Reset. The Start button is used to activate the system, the Stop button is used to deactivate it, and the Reset button is used to clear faults and reinitialize the system when needed.

In addition, the HMI includes two level indicators: a high-level indicator and a low-level indicator. These indicators provide visual feedback when the tank level exceeds the upper limit or drops below the lower limit.

The system also includes three status indicators corresponding to the Start, Stop, and Reset conditions, allowing the operator to easily identify the current state of the system.

Overall, the HMI provides a simple and effective interface for controlling the system and monitoring its basic operating conditions.

7 - Results and Performance Analysis

7.1 System Response

The system successfully maintains the tank level at the desired setpoint. The response shows that the level increases or decreases smoothly until it reaches the required value.

7.2 Stability and Accuracy

The system operates with acceptable stability and minimal oscillations. However, during testing, it was observed that the fine tuning process did not produce accurate or consistent results. Therefore, the pretuning method was used, which provided more stable and reliable performance for the system.

7.3 Controller Performance

The PI controller effectively reduces the error and maintains steady operation. Using the pretuned parameters resulted in a faster and more stable response compared to the fine tuning approach.

8 - Conclusion

8.1 Summary

A tank level control system was successfully designed and implemented using PLC programming in TIA Portal and simulated using Factory I/O and PLCSIM. The system uses a PI controller to regulate the liquid level based on a closed-loop control strategy.

8.2 System Performance

The system achieved stable and accurate control of the tank level. The pretuning method provided reliable controller parameters, resulting in smooth response and minimal steady-state error. Safety conditions and control logic ensured proper system operation under different scenarios.

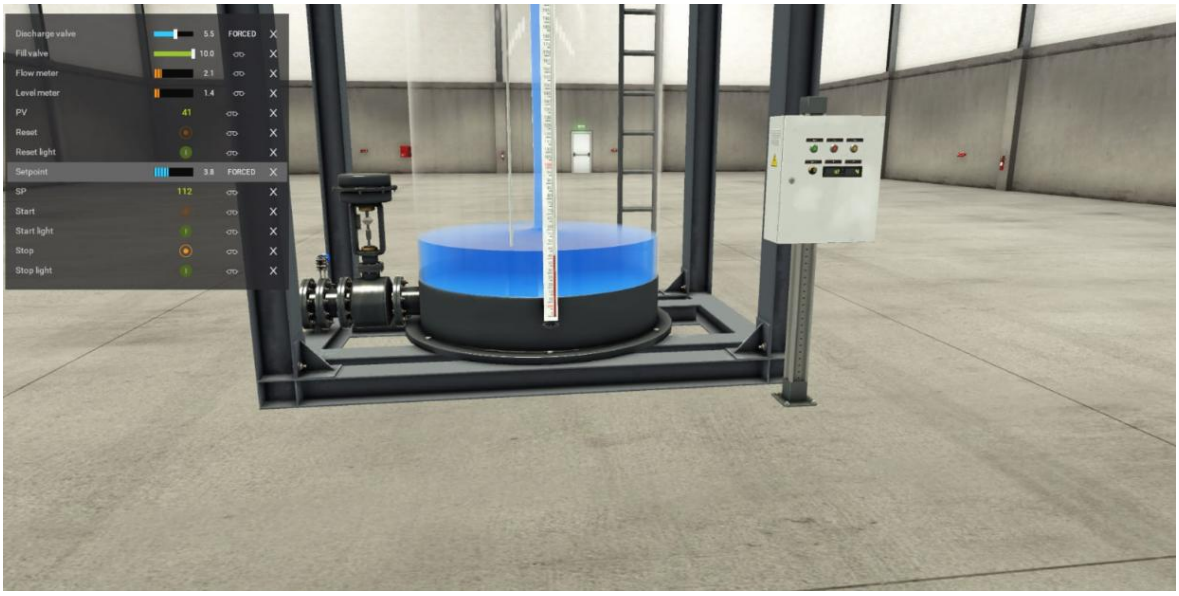
8.3 Final Remarks

The project demonstrates the effectiveness of PLC-based control systems in industrial applications. It also highlights the importance of proper controller tuning and system design to achieve reliable and efficient performance.

9 - Team Contribution

Member	Contribution
Kareem Ahmed	PLC programming and implementation of LAD logic
Ahmad Hatem	PID controller setup and tuning
Mahmoud Karam	HMI

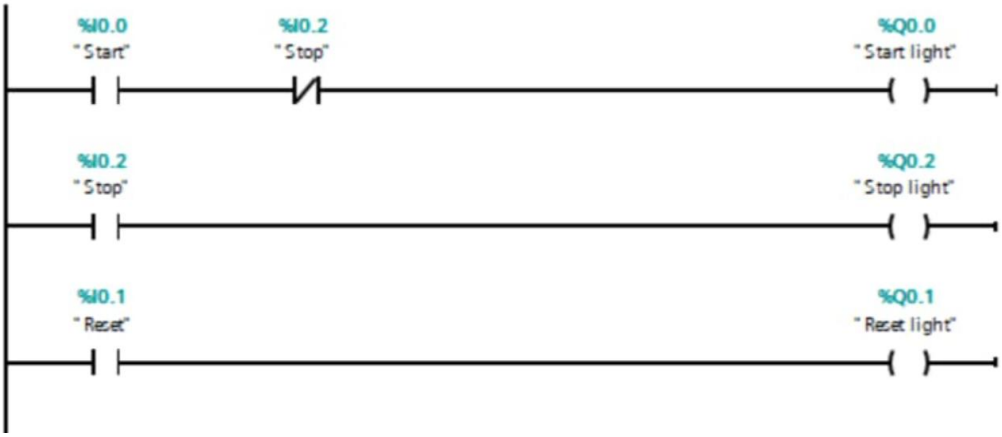
10 – APPENDICES





Network 6: Start/Stop Logic

This Network shows the start and stop logic for classic control for the whole system



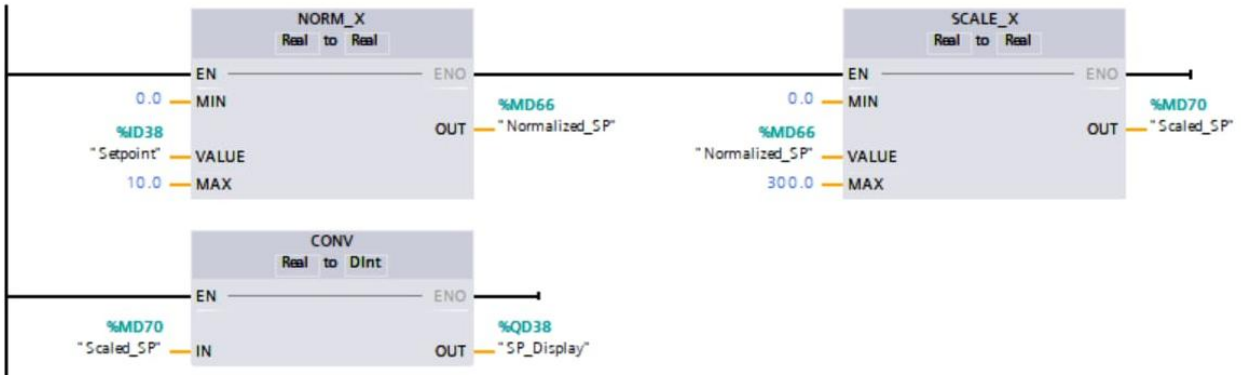
Network 4:

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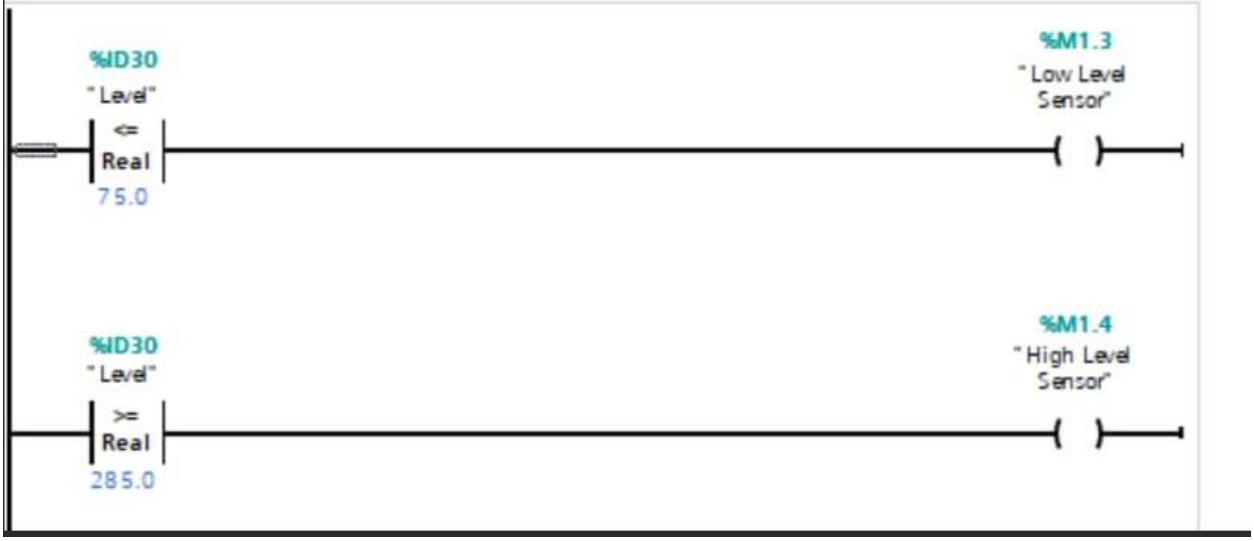
Network 3:

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Network 5: Comparator Functions

This Networks Uses Comparator Functions to compare the values of the currenc t water l...



Network 2:

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